BAAO COMMUNITY COLLEGE
SAN JUAN BAAO CAMARINES SUR

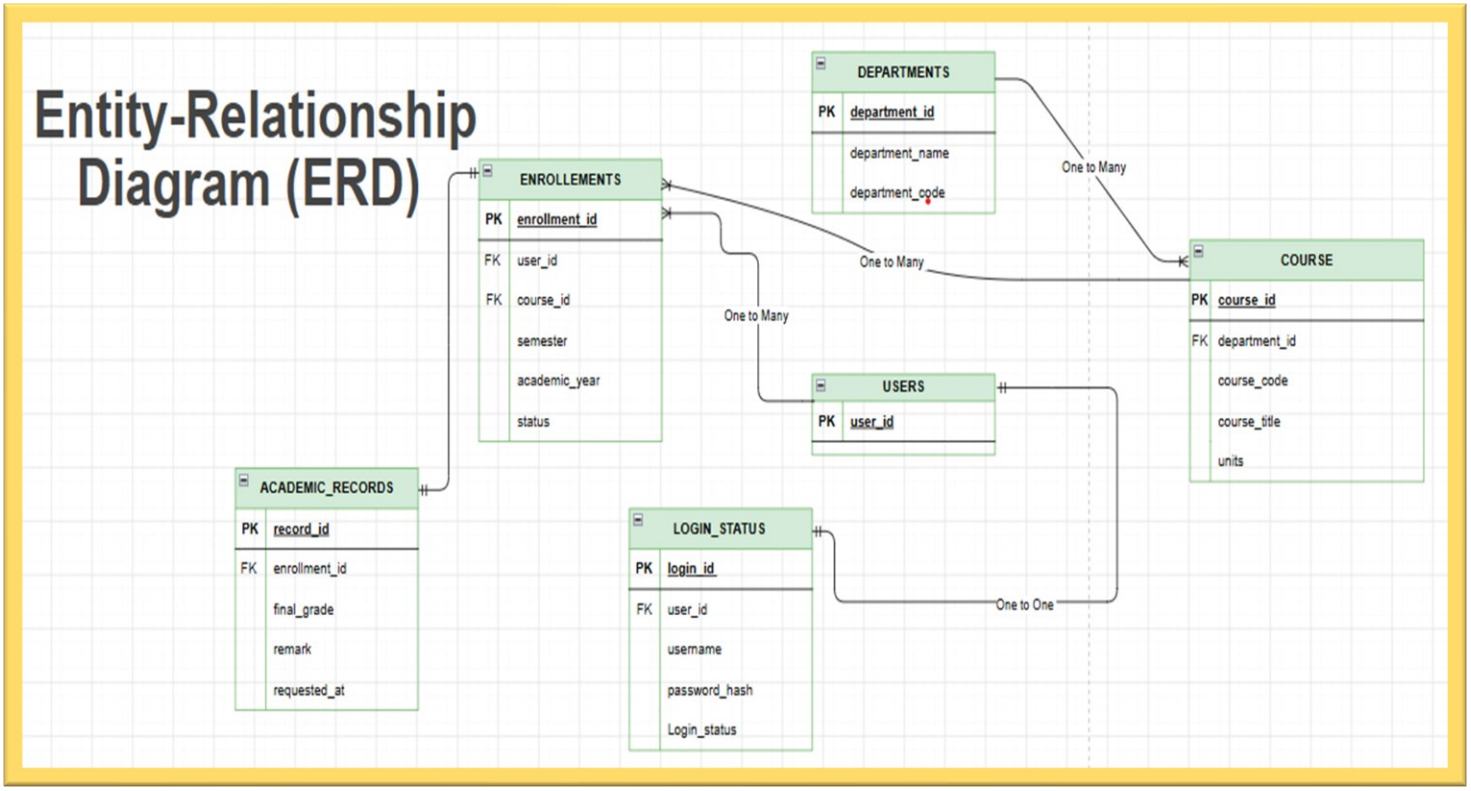
FINAL PROJECT IN CC105-INFORMATION MANAGEMENT

GROUP 4: SYNTACTIC SYNDICATE

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ENTITY RELATIONSHIP DIAGRAM (ERD)



DATA DICTIONARY

DEPARTMENTs Table

Column Name	Data Type	Key Type	Description
department_id	INT	**PK**	Unique identifier for the department.
department_name	VARCHAR		Full name of the department.
department_code	VARCHAR		Short code for the department.

COURSE Table

Column Name	Data Type	Key Type	Description
course_id	INT	**PK**	Unique identifier for the course.
department_id	INT	**FK**	References the department offering the course.
course_code	VARCHAR		Official course code (e.g., CC 106).
course_title	VARCHAR		Descriptive name of the course.
units	INT/DECIMAL		Credit units assigned to the course.

USERs Table

Column Name	Data Type	Key Type	Description
user_id	INT	**PK**	Unique identifier for the user profile.

LOGIN_STATUS Table

Column Name	Data Type	Key Type	Description
login_id	INT	**PK**	Unique identifier for the login record.
user_id	INT	**FK**	References the parent user profile.
username	VARCHAR		Unique login handle.
password_hash	VARCHAR		Securely hashed user password.
Login_status	VARCHAR/ENUM		Current login status indicator.

ENROLLMENTs Table

Column Name	Data Type	Key Type	Description
enrollment_id	INT	**PK**	Unique identifier for the enrollment transaction.
user_id	INT	**FK**	References the enrolled user/student.
course_id	INT	**FK**	References the enrolled course.
semester	VARCHAR		Academic semester of registration.
academic_year	VARCHAR		Academic year period.
status	VARCHAR/ENUM		Current status of the enrollment.

ACADEMIC_RECORDs Table

Column Name	Data Type	Key Type	Description
record_id	INT	**PK**	Unique identifier for the academic record.
enrollment_id	INT	**FK**	References the specific enrollment instance.
final_grade	DECIMAL		Recorded final grade score.
remark	VARCHAR		Standing remarks (e.g., Passed, Failed).
requested_at	TIMESTAMP		Timestamp of grade release or record request.

Database Normalization Proof (3NF)

This database structure complies fully up to the Third Normal Form (3NF) to ensure optimal efficiency and data integrity

1. First Normal Form (1NF):

- All tables contain only atomic values with no repeating groups or multi-valued fields. Each attribute holds a single value, and every table has a defined primary key (department_id, course_id, user_id, login_id, enrollment_id, and record_id).

2. Second Normal Form (2NF):

- The database meets 1NF criteria. Furthermore, all tables utilize single-column primary keys, completely eliminating any possibility of partial functional dependencies where a non-key attribute depends on only part of a composite key.

3. Third Normal Form (3NF):

- The database meets 2NF and contains no transitive dependencies (no non-key attribute depends on another non-key attribute).
- For instance, login credentials (username, password_hash) are isolated in login_status rather than cluttering users.
- Similarly, grading details are separated into academic_records, referencing enrollments via a foreign key (enrollment_id), ensuring that operational registration data and performance evaluation metrics are cleanly decoupled.